

HTTP

HTTP defines requests and responses between a client and a server. A request contains a method, target, headers, and sometimes a body. A response contains a status code, headers, and optionally a body. HTTP itself does not encrypt the message. In this course, HTTP is a prerequisite for understanding TLS in web communication, and TLS is a prerequisite for understanding HTTPS. Learn this sequence: HTTP, then TLS, then HTTPS.

TLS

TLS protects data in transit using encryption and integrity checks. A certificate helps the client authenticate the server during the handshake. A browser checks that the certificate matches the requested hostname and chains to a trusted authority. TLS does not decide whether a signed-in user can read a particular document. HTTP is a prerequisite for studying TLS in this web course; TLS is a prerequisite for HTTPS.

HTTPS

HTTPS carries HTTP messages over TLS. To understand HTTPS, study TLS first and HTTP before TLS. This makes HTTP a foundational prerequisite two steps before HTTPS. HTTPS protects traffic between browser and server against passive reading and modification in transit. It does not repair application authorization bugs, prevent a malicious server from storing data, or guarantee that a website is honest.

DNS

DNS resolves domain names to records such as IP addresses. A resolver can reuse a cached answer until its time to live expires. A DNS lookup does not prove that an application is healthy. After locating a server, the browser still needs a successful connection and an HTTP response. DNS caching and HTTP caching operate at different layers: one reuses name resolution results, the other reuses application responses.

HTTP Caching

HTTP caching reuses a stored response when its freshness rules permit.

Cache-Control can specify private responses or prohibit storage with no-store. An

ETag is a validator that can support conditional requests and avoid downloading an

unchanged representation. Never assume a shared cache may safely store a

personalized response. This response cache differs from a database index, which

accelerates locating rows without storing an HTTP response.

Status Codes

An HTTP 401 response means authentication is required or the supplied credentials are not accepted. An HTTP 403 response means the request is forbidden; it can occur even when identity is known. HTTP 429 indicates too many requests and can include a Retry-After header. HTTP 503 indicates that a service is temporarily unavailable. Clients should distinguish these cases rather than report every failure as an invalid password.